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Q4. Network Layer Case Study: MTU Path Discrepancy and Performance Analysis of TCP, UDP, HTTP and DNS

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Executive Summary:

In this case study, users of a web application can successfully load the login page, but the browser hangs when the Submit button is clicked. Small network packets pass through the firewall successfully, while larger packets fail. Investigation shows that the communication path contains a VPN tunnel with an effective Maximum Transmission Unit (MTU) of 1300 bytes.

The case study also compares the behavior of **TCP, UDP, HTTP, and DNS** under normal and reduced MTU conditions. TCP can recover from packet loss through retransmission, while UDP does not provide retransmission. HTTP, which normally operates over TCP, can experience delays or hanging connections when TCP packets are repeatedly lost. DNS normally uses UDP for small queries but may use TCP when responses are large.

The recommended solution is to allow the required ICMP messages through the firewall and correctly configure PMTUD. This allows devices to determine the appropriate packet size and prevents large packets from being unnecessarily dropped.

Problem Statement & Topology:

A company uses a VPN connection to access a web application hosted on a remote network.

Users report the following symptoms:

- The website login page loads successfully.
- Clicking Submit causes the browser to hang.
- Small ping packets work successfully.
- Large packets fail.
- The VPN tunnel has an effective MTU of 1300 bytes.
- The firewall blocks important ICMP messages.

The problem occurs because packets larger than the path MTU cannot travel through the VPN tunnel.

Technical Analysis & Troubleshooting

What is MTU?

Maximum Transmission Unit (MTU) is the largest packet size that can be transmitted over a network interface without fragmentation.

Test using small packet:

```
C:\Windows\System32>ping 8.8.8.8 -f -l 1200

Pinging 8.8.8.8 with 1200 bytes of data:
Reply from 8.8.8.8: bytes=1200 time=84ms TTL=113
Reply from 8.8.8.8: bytes=1200 time=54ms TTL=113
Reply from 8.8.8.8: bytes=1200 time=70ms TTL=113
Reply from 8.8.8.8: bytes=1200 time=60ms TTL=113

Ping statistics for 8.8.8.8:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 54ms, Maximum = 84ms, Average = 67ms
```

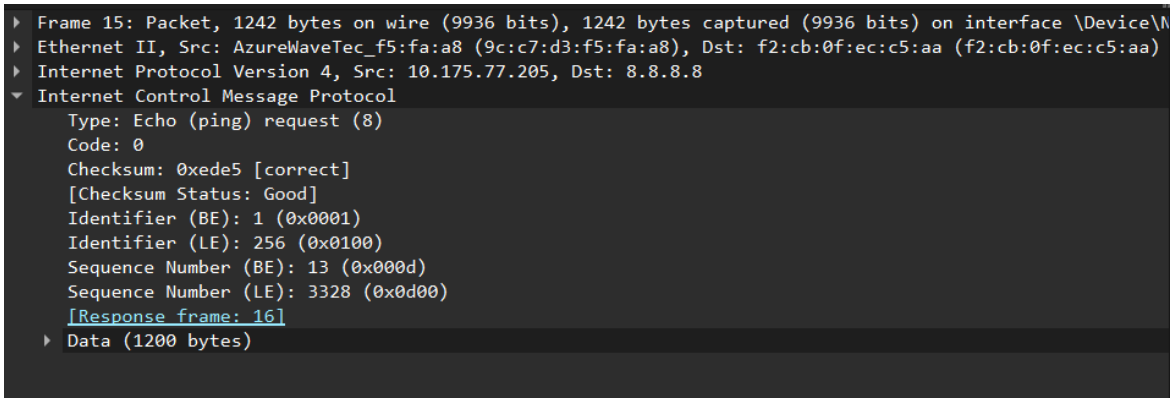
Test using large packet:

Using large packets taking 75% loss of packet

```
C:\Windows\System32>ping 8.8.8.8 -f -l 1400

Pinging 8.8.8.8 with 1400 bytes of data:
Reply from 10.175.77.18: Packet needs to be fragmented but DF set.
Packet needs to be fragmented but DF set.
Packet needs to be fragmented but DF set.
Packet needs to be fragmented but DF set.

Ping statistics for 8.8.8.8:
    Packets: Sent = 4, Received = 1, Lost = 3 (75% loss),
```



Protocol	Normal MTU	Reduced MTU	Effect of Packet Loss
TCP	Normal communication	Retransmissions and reduced throughput	TCP retransmits lost segments
UDP	Normal communication	Large datagrams may be fragmented/dropped	No built-in retransmission
HTTP	Webpage loads normally	Requests/responses may become slow or hang	TCP retransmission affects HTTP
DNS	Small queries work	Large responses may fail	DNS can retry or use TCP

Proposed Solution & Implementation Plan:

The primary solution is to allow the required ICMP messages through the firewall so that Path MTU Discovery can work correctly.

For IPv4:

ICMP Type 3, Code 4 — Destination Unreachable / Fragmentation Needed

Changing MTU or firewall rules can temporarily affect network connectivity.

Possible risks include:

- Temporary VPN disconnection
- Reduced network performance if MTU is set incorrectly
- Incorrect firewall rules affecting security
- TCP connections being interrupted during configuration

Fallback Plan

Before making changes:

1. Record the existing configuration.
2. Change one parameter at a time.
3. Test connectivity after each change.
4. Keep a backup of the router/firewall configuration.

Conclusion:

The case study demonstrates how an MTU mismatch in a VPN tunnel can cause seemingly unrelated application problems. Although small packets successfully pass through the network, larger packets can be dropped when the effective MTU is only 1300 bytes.

TCP attempts to recover from packet loss using retransmission, while UDP does not provide such recovery. HTTP performance is affected because it relies on reliable transport, and DNS may be affected when responses become large.

The main solution is to ensure that Path MTU Discovery works correctly by allowing ICMP Type 3, Code 4 messages through the firewall, together with appropriate MTU or TCP MSS configuration.

This analysis demonstrates that understanding the interaction between network-layer behavior and TCP, UDP, HTTP, and DNS performance is essential for diagnosing real-world network failures.

References:

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 RFC 9110 – HTTP Semantics, IETF, 2022.
RFC 9110

 RFC 9000 – QUIC: A UDP-Based Multiplexed and Secure Transport, IETF, 2021.
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